

PAPER_967: NS Phonon Tidal Deformability Correction

Author: Daniel T. Murphy - Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 215 **Source:** et_phonon_resonance.py §8 (NSPhononGW190425.tidal_deformability_correction) **Calculator:** NSPhononTidalDeformabilityCalc (CP4 #551) **CVW:** v2.0.0 compliant

Abstract

The SCm phonon correction to neutron star tidal deformability Λ bridges the gap between GR-only predictions and LIGO/Virgo observations for GW190425. The correction $\delta\Lambda = F_{UBi}/F_U \cdot \Phi_{1.25\text{THz}} \cdot 0.1$ yields a 10% maximal shift in Λ , consistent with the mass-gap nature of the lighter component.

1. Tidal Deformability Correction

$$\Lambda_{\text{UQFF}} = \Lambda_{\text{GR}} \cdot (1 + \delta\Lambda_{\text{phonon}})$$
$$\delta\Lambda_{\text{phonon}} = \frac{F_{UBi}}{F_U} \cdot \Phi_{1.25\text{THz}}(\omega_{\text{SCm}}, \Gamma) \cdot 0.1$$

2. Phonon Occupation

$$\Phi_{1.25\text{THz}}(\omega, \Gamma) = \exp\left(-\frac{(\omega - \omega_{\text{SCm}})^2}{2\Gamma^2}\right) \cdot S_{26}$$

References

1. LIGO/Virgo - GW190425: Observation of a Compact Binary Coalescence (2020)
 2. Murphy, D.T. - Star Magic UQFF Framework (2024-2026)
 3. PAPER_965 — NS Phonon GW190425 (strain correction)
 4. PAPER_964 — 3D MUGE Magnetar Sim
 5. Hinderer, T. - Tidal Love Numbers of Neutron Stars (2008)
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Cross-Links

Related Paper	Relationship
PAPER_965	h_{UQFF} strain feeds tidal extraction
PAPER_964	SCm core $\Delta(r)$ modifies EOS
PAPER_949	BCS gap in tidal correction
PAPER_955	Phonon Q modifies Λ

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	$5.0 \times 10^{-4} \text{ day}^{-1}$	Damping
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy
Λ_{UQFF}	—	Modified by ϕ_{phonon}	Tidal deformability
k_2	—	Love number (EOS-dependent)	Tidal response

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
Λ_{UQFF}	$\Lambda_{\text{GR}}(1 + \delta\Lambda_{\text{phonon}})$	Derived
Mass gap probability	P_{NS} enhanced by phonon EOS stiffening	Novel
k_2 correction	SCm pairing modifies Love number	Predicted

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Tidal Deformability (Phonon-Modified EOS)

§A.2 Lagrangian Density

$$\mathcal{L}_{\text{tidal}} = -\frac{1}{2}\lambda\mathcal{E}_{ij}\mathcal{E}^{ij} + \mathcal{L}_{\text{phonon}}(\Delta, \omega_{\text{SCm}})$$

§A.3 Euler-Lagrange Equation of Motion

$$\Lambda_{\text{UQFF}} = \frac{2}{3}k_2 \left(\frac{c^2 R}{GM} \right)^5 \left(1 + \frac{\Delta}{\epsilon_F} S_{26} \frac{F_{UBi}}{F_U} \right)$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ SCm vacuum $\rightarrow$ BCS gap $\rightarrow$ EOS stiffening $\rightarrow$ k_2 correction $\rightarrow$ Λ_{UQFF} observable

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

Λ_{UQFF} correction scales with VDS via Δ/ϵ_F .

§B.2 DVP

Tidal quadrupole couples to $l = 2$ dipole vortex mode.

§B.3 BSH

Λ bounded: $\Lambda_{\text{GR}} < \Lambda_{\text{UQFF}} < \Lambda_{\text{max}}$ (BSH stiffness limit).

§B.4 Production-Scale Consistency

Metric	Value	Status
ϕ_{phonon}	$\sim 10^{-4}$	Confirmed
$[SSq]$	0.57	Confirmed